Rafsun Jany Arman

armanrafsunjany@gmail.com | rafsunjanyarman.com | kaggle.com/rafsun/code | github.com/rjarman
linkedin.com/in/rafsun-jany-arman | medium.com/@rafsunjanyarman | researchgate.net/profile/Rafsun-Jany-Arman

Experience

Senior Software Engineer Nov. 2025 – Present
SELISE Group *Dhaka, Bangladesh (Hybrid)*

- Architected a secure, on-premise **Enterprise Generative AI Ecosystem** running two production engines: a deterministic chat pipeline and an autonomous deep-agent runtime on **LangGraph**, with checkpointed conversations, human-in-the-loop interrupts, and real-time streaming to concurrent **WebSocket** connections.
- Engineered a production **RAG** platform on **Qdrant** with hybrid dense+sparse (BM25) retrieval, **Cohere Rerank v4** reranking, and queue-based ingestion powered by **Docling** multi-format parsing (PDF, Office, tables, OCR) and web crawling.
- Built a dynamic **MCP** tool platform converting any business's OpenAPI/Swagger spec into agent tools at runtime, with capacity-aware meta-tool collapse, per-user OAuth, reactive mid-conversation auth, and **Composio** integration for third-party SaaS actions.
- Built **event-driven autonomous agents**: **Composio** triggers send signed webhooks that are verified, deduplicated, and queued into durable agent runs, so agents act on events from Gmail, Slack, GitHub and other apps without a user prompt. Added user-defined **subagents** that the main agent delegates work to, with token usage tracked per subagent.
- Implemented security-first agent design: severity-based guardrails, sandboxed per-user code execution, encrypted per-user tool credentials, and organization-level data isolation enforced at the database boundary.
- Instrumented the full agent stack with **Opik** LLM tracing and per-step latency/token analytics, enabling production debugging of multi-step agent runs and per-conversation cost attribution.
- Developed a high-concurrency **Collaborative Chat Engine** with **FastAPI**, where AI agents join multi-user rooms as first-class participants with **Redis**-backed run queuing, resumable streams, and agentic web search (**Perplexica**, **Tavily**). Extended it with an embeddable public chat widget secured by short-lived tokens and origin allowlists.

Software Engineer Nov. 2023 – Oct. 2025
SELISE Group *Dhaka, Bangladesh (Remote)*

- Developed the all-in-one **Ski World Cup App** using **Flutter** and **Bitmovin**, delivering sub-3s latency live streaming and real-time athlete telemetry for over 30k users.
- Built a dynamic **Maritime Data Visualization Dashboard** for a global marine services client, enabling predictive alerts and efficient data filtering across multiple operational systems.
- Engineered a scalable **IoT Middleware** using **Java Spring Boot**, **RabbitMQ**, and **TimescaleDB** to orchestrate millions of telemetry messages for Smart Metering and Industrial use cases.
- Designed end-to-end **Smart Metering Integration** solutions, automating data collection from diverse hardware protocols to optimize utility consumption analysis.

Junior Software Engineer Aug. 2022 – Oct. 2023
SELISE Group *Dhaka, Bangladesh (Hybrid)*

- Extended an **IoT Commodity Tracking Platform** for a leading mining company, ensuring accurate shipment monitoring and bag condition tracking across international ports.
- Created a **Production Cost Monitoring System** for a Swiss food manufacturer, utilizing IoT data to track electricity usage and streamline operational efficiency, resulting in reduced waste.
- Enabled **White-Label IoT Solutions** by developing customizable branding tools and rule chains, allowing B2B clients to tailor platforms to specific operational branding needs.

Intern Software Engineer Mar. 2022 – Jul. 2022
SELISE Group *Dhaka, Bangladesh (Remote)*

- Conducted R&D on **ThingsBoard IoT** platform features and designed custom D3.js data visualization graphs to meet specific client reporting requirements.
- Published an NPM package for file uploads in **NativeScript** (Android API ≥ 30) and identified security vulnerabilities in internal systems, earning recognition.

Research

Fish classification using saliency detection | *Computer Vision, U2-Net, HOG*

- Developed a fish classification system utilizing saliency detection (depending on shape and texture) and ensemble learning (KNN, SVM, etc.), achieving up to 100% accuracy on local fish datasets.

Fish Observatory and Census System (FOCS) | *Image Processing, Research*

- Proposed and implemented a census system for automated fish monitoring and population estimation using advanced image processing techniques.

Parkinson's Disease Detection from Handwriting | *Deep Learning, Evaluation, On-Device AI* Ongoing

- Audited 16 published studies on spiral and wave drawing datasets and found 13 split data by image instead of by participant, which leaks the same person into train and test. Re-ran my own earlier 96.67% model with participant-grouped splits, where it dropped to about 76–84%, and reported the corrected number.
- Ran 20 controlled experiments including cross-dataset tests (HandPD, NewHandPD), fixed a Grad-CAM bug, and deployed the model on-device with **ExecuTorch** with Grad-CAM computed inside the exported graph, benchmarked on 3 phones. Manuscript in preparation.

Bangla & **Bus-Mama** | *NLP, Python, Mobile Dev*

- Created 'Bangla', a Python library for Bengali text processing and lemmatization using Trie data structures.
- Applied the library in 'Bus-Mama', a real-time tracking application with a Bengali-language chatbot for university students.

Projects

CrewsBoard | *Agentic AI, QA Automation* In Development

- Building an agentic QA system that tests every pull request against its acceptance criteria. An agent runs browser tests and a separate verifier agent judges the evidence.
- Code never leaves the user's machine: a local service runs each app in an isolated sandbox, and fixes ship as separate PRs only after human approval.

LiveKit Voice Agent | *Python, LiveKit, LangGraph, Whisper, MCP*

- Built a self-hosted AI voice/video stack with **LangGraph** orchestration, **Whisper** STT, real-time audio handling, and **MCP** tool integration for production agentic voice workflows.

Adaptive Marketing AI | *Python, LangChain, AI/ML, Marketing API*

- Built an intelligent marketing engine that interfaces with multiple ad platforms to generate and deploy targeted campaigns based on data-driven insights.

AI Brand Mention Tracker | *TypeScript, Firestore, LLM Observability*

- Built a backend service analyzing how a brand surfaces in AI-generated responses, with scheduler-driven analysis, score-change alerts, and Firestore-based event triggers.

Technical Skills

Languages: Python, TypeScript, SQL, Java
Frameworks: FastAPI, React
AI / LLM Stack: LangGraph, LangChain, deepagents, CrewAI, Qdrant, MCP (Model Context Protocol), Composio, Opik (LLM Observability), LiveKit, Whisper, Perplexica, Tavily
Data & Infra: PostgreSQL, TimescaleDB, MongoDB, Redis, Firestore, RabbitMQ, Docker
Concepts: Production Agentic AI, RAG, Multi-Agent Systems, On-Premise GenAI, AI Guardrails & Agent Security, AI-First Development (Claude Code), Microservices, IoT, System Architecture, CI/CD

Certifications & Awards

Claude Certified Developer, Foundations | *Anthropic* 2026
Language & Knowledge Engineering Lab | *8th Int. Symposium (LKE 2021)* 2021
Top 100 | *Idea Contest Mujib 100* 2021
Honorable Mention | *CanSat Competition (BRAC)* 2019
Trainer | *Mechatronics Club, GSTU* 2019

Education

Gopalganj Science and Technology University (GSTU) Gopalganj, Bangladesh
Bachelor of Science in Computer Science & Engineering, CGPA: 3.41 *Jan. 2017 – May 2022*